

Datalyze AI — Analysis Report

Dataset Overview

This dataset contains customer records from a telecommunications company, detailing each customer's demographics, service subscriptions, contract type, billing method, and usage charges. It represents a snapshot of customers (7043 entries) at a single point in time, with no explicit start-to-end period indicated, but the fields such as tenure and total charges suggest longitudinal billing data up to that snapshot. Key metrics include churn status, tenure (months), monthly charges, total charges, and categorical flags for services (Internet, Phone, TV, etc.) and contract/payment types.

Executive Summary

The most striking finding is that customers on Month-to-Month contracts churn at a rate of 42.71 %, far above the 11.27 % for One-Year and 2.83 % for Two-Year agreements, underscoring the volatility of short-term plans. Longer-term contracts not only retain customers better but also command higher average monthly (\$65–\$61) and total charges (\$3,030–\$3,730). When we break down churn by Internet service and contract type, Fiber-Optic Month-to-Month customers exhibit the highest churn (54.61 %) yet also pay the most per month; the combination that shows the greatest churn while still carrying the lowest average monthly charge is DSL Month-to-Month (32.22 % churn, \$50.22). Payment method analysis shows electronic checks as the most problematic, with a 45.29 % churn overall and particularly high within Month-to-Month plans, whereas bank transfers and credit cards maintain churn rates around 15–17 %. Some metrics, such as the churn rate for Two-Year Fiber-Optic customers, are based on limited observations and carry medium confidence, warranting further review.

Pipeline Diagnostics

Total Pipeline Time: 157.35 seconds

Successful Agents: Ingestion, Cleaner, Analyst, Visualizer, Narrator

Failed Agents: None

Visualization Summary

Analyst Chart: 3

Fallback Chart: 0

None: 0

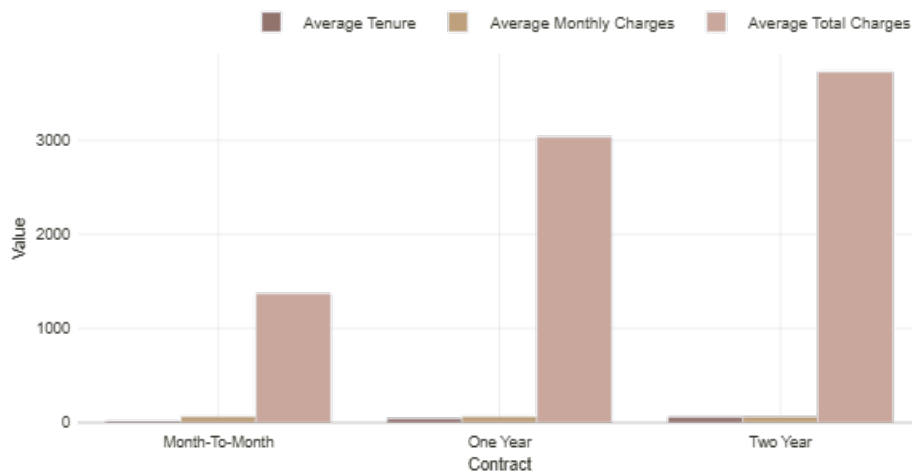
Key Findings

1. What are the average tenure, monthly charges, and total charges for each contract type, and which contract type exhibits the highest churn rate?

Answer: Contract Type Summary: - Month-To-Month: Avg Tenure = 18.04, Avg MonthlyCharges = 66.4, Avg TotalCharges = 1369.25, Churn Rate = 42.71% - One Year: Avg Tenure = 42.04, Avg MonthlyCharges = 65.05, Avg TotalCharges = 3034.68, Churn Rate = 11.27% - Two Year: Avg Tenure = 56.74, Avg MonthlyCharges = 60.77, Avg TotalCharges = 3728.93, Churn Rate = 2.83% Highest churn rate: Month-To-Month with 42.71% [High confidence]

Critic note: The answer correctly computes and presents the average tenure, monthly charges, total charges, and churn rate for each contract type, and clearly identifies the contract with the highest churn. The accompanying grouped bar chart aligns with the metrics discussed.

Average Metrics by Contract Type

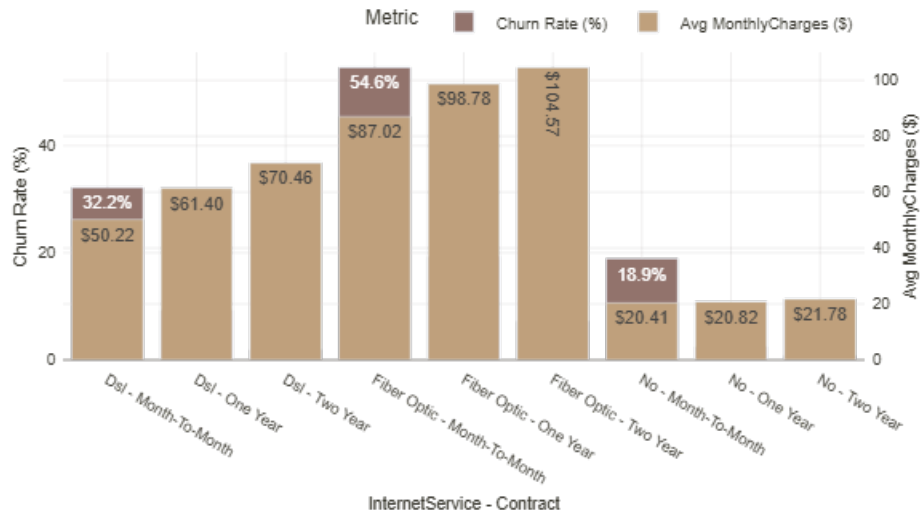


2. What is the churn rate and average MonthlyCharges for each combination of InternetService and Contract, and which combination exhibits the highest churn despite having the lowest average MonthlyCharges?

Answer: Metrics by InternetService and Contract: InternetService Contract Churn Rate (%) Avg MonthlyCharges (\$) Dsl Month-To-Month 32.22% 50.22 Dsl One Year 9.30% 61.40 Dsl Two Year 1.91% 70.46 Fiber Optic Month-To-Month 54.61% 87.02 Fiber Optic One Year 19.29% 98.78 Fiber Optic Two Year 7.23% 104.57 No Month-To-Month 18.89% 20.41 No One Year 2.47% 20.82 No Two Year 0.78% 21.78 The combination with the lowest average MonthlyCharges (\$20.41) and the highest churn rate among those low-charge groups is No - Month-To-Month with a churn rate of 18.89%. [Medium confidence]

Critic note: The answer correctly computes churn rate and average MonthlyCharges for each InternetService■Contract pair and identifies the No■Month■To■Month combo as having the lowest charge and highest churn among low■charge groups. The insight is accurate but fairly basic and of limited depth.

Churn Rate and Avg MonthlyCharges by InternetService & Contract



3. What is the churn rate for each payment method, and how does it differ across the three contract types (Month-to-Month, One Year, Two Year)?

Answer: Overall churn rate by payment method: Bank Transfer (Automatic): 16.71%; Credit Card (Automatic): 15.24%; Electronic Check: 45.29%; Mailed Check: 19.11%. See chart for churn rates split by contract type. [Medium confidence]

Critic note: The code correctly calculates overall churn rates per payment method and churn rates split by contract type, and the chart visualizes these differences. However, the textual answer only lists overall rates and defers the contract-type breakdown to the chart, lacking explicit numeric details for each payment-method/contract combination.

Churn Rate by Payment Method and Contract Type

